

SOUMYA RANJAN BEHERA

Lead AI Engineer | GenAI Architect | LLM & RAG Systems

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PROFESSIONAL SUMMARY

8+ years in software and AI engineering, with the last few years focused on **agentic AI systems, LLM orchestration, and GenAI products** in production. Currently at JPMorgan Chase building **ChatCFO** — an agentic RAG system where a CFO can query live finance data through a multi-step reasoning pipeline built on an in-house LLM, hybrid retrieval (FAISS + BM25), and a full orchestration layer. Before that, three years at Siemens building **Industrial CoPilot** — a Semantic Kernel agent orchestrator with composable plugins — and **DomainGenie**, a conversational AI agent that reached 10 teams across 4 countries and supported migration of ~3 million lines of legacy code.

Good at the full stack of an AI system: picking the right retrieval strategy, wiring up the inference layer, keeping latency and cost in check, and making sure the outputs are explainable enough to satisfy compliance teams. Equally comfortable writing the architecture doc and doing the implementation.

CORE TECHNICAL SKILLS

LLMs & GenAI:	Agentic AI, Agent Orchestration, RAG (FAISS, BM25, Hybrid), Prompt Engineering, Embeddings, Contextual Memory, Conversation History, Semantic Kernel, Plugins, Parallel Execution
AI Frameworks:	LangChain, LlamaIndex, Hugging Face, Streamlit, GCP Dialogflow, Azure Cognitive Services, OpenAI API
Languages:	Python, C# / .NET 8, ASP.NET Web API, REST APIs, SQL, JavaScript
Data & Cloud:	Databricks (DBX), PySpark, SQL Server, SSIS, ETL Pipelines, Azure (AI-900 certified), GCP, Parquet
Engineering:	Microservices, CI/CD, SOLID Principles, Design Patterns, Unit Testing, System Architecture

PROFESSIONAL EXPERIENCE

Senior Quant Modelling Engineer • Lead AI Architect · [JPMorgan Chase & Co.](#) May 2025 – Present · Bengaluru

- **Agentic AI architecture lead** for ChatCFO — designed the full agent orchestration layer including multi-step reasoning, tool routing, and dynamic context assembly for CFO and finance workflows in a tightly regulated environment.
- Built the **prompt orchestration pipeline** handling contextual memory, conversation history, and parallel execution — so the agent maintains coherent multi-turn dialogue across complex finance queries without losing context.
- Retrieval layer uses **hybrid FAISS + BM25** search over Parquet-based structured finance data, feeding clean, ranked context into the agent at inference time.
- Made sure every agent response is **explainable and auditable** — a non-negotiable in this environment. Worked closely with finance and compliance stakeholders to get that right.
 - ▶ **ChatCFO** — *Agentic AI · LLM Orchestration · FAISS · BM25 · Databricks · Python · Prompt Engineering*
 - End-to-end agentic pipeline: in-house LLM, **agent orchestration**, tool routing, hybrid vector + keyword retrieval (FAISS + BM25), content extraction, contextual memory, conversation history, parallel execution, and Databricks data retrieval for live finance queries.
 - **80% improvement** in daily advisor contributions and feedback loop performance after rollout.

Senior Software Engineer – AI & Platform Engineering · [Siemens Technology & Services](#) Jun 2022 – May 2025 · Bengaluru

- Worked on **agentic AI systems** within Siemens' industrial automation stack — building Semantic Kernel-based agent orchestrators, plugins, and LLM pipelines for the TIA Portal ecosystem.

- Owned the **Translation Module in Industrial CoPilot**: plugged into the Semantic Kernel orchestration layer as a standalone agent plugin — handling NLP + LLM semantic translation of project texts across domain-specific industrial terminology.
 - For DomainGenie, designed the **full conversational agent** on GCP Dialogflow — intents, entities, training phrases, and document knowledge base — then rolled it out to developer teams across Europe and Asia.
 - Also built the GUI Analyzer: a Roslyn-based static code analyzer integrated into the CI/CD pipeline to enforce HDPI compliance at commit time.
- ▶ **Industrial CoPilot** — *Agentic AI · Semantic Kernel · Plugins · Orchestration · In-house LLM · RAG · .NET 8 · ASP.NET Web API*
 - **Semantic Kernel agent orchestrator** driving TIA Portal feature automation — plugin architecture with discrete, composable agent plugins handling different automation tasks. My module was the **Translation Plugin**: NLP + LLM semantic translation pipeline for project texts and domain content, registered as a native Semantic Kernel plugin within the orchestration layer.
 - ▶ **DomainGenie** — *GCP Dialogflow · Conversational Agent · Python · C# · Streamlit · RAG · Embeddings*
 - **Conversational AI agent** on GCP Dialogflow — designed the full agent (intents, entities, fulfillment, document RAG) to help legacy .NET developers navigate HDPI migration. Reached **10 teams across 4 countries**, supporting migration of **~3 million lines of code** across **16 transformation rules** and 7 major UI patterns.
 - ▶ **GUI Analyzer (HDPI Code Analyzer)** — *C# · Roslyn API · CI/CD · GIT · Unit Testing*
 - Static analyzer built with the Roslyn API to catch HDPI compliance violations in .NET codebases — wired into CI/CD so issues flagged at commit time, not during review.

Senior Software Engineer · Tech Mahindra

May 2021 – Jun 2022 · Hyderabad

- Delivered backend systems and data extraction tooling for enterprise clients — primarily **API-centric architectures and ETL pipelines**. One automation project cut manual effort by **70%**.
- Involved early in solution scoping with clients, which helped keep delivery timelines realistic — **~50% improvement in deliverable speed** compared to previous project cycles.
- Ran architecture sessions and contributed to planning discussions, mostly around keeping the backend maintainable as requirements changed mid-project.

Software Engineer · L&T Technology Services

Oct 2017 – May 2021 · Vadodara / Mysore

- Four years building data-driven applications and automation tools for industrial clients — ETL pipelines, document extraction systems, and Windows applications for internal operations.
 - Automation work reduced manual effort by **50%** on data processing workflows; reusable infrastructure components improved subsequent development effort by **40–50%** on follow-on projects.
- ▶ **Ainfonyx** — *Azure Cognitive Services · OCR · Angular · ASP.NET MVC · Web API · SQL Server*
 - Web app for automated document processing using **Azure Cognitive Services** and Abbyy OCR — handled file property extraction and digitisation across multiple document formats and input types.

CERTIFICATIONS

Microsoft Azure AI-900 — Azure AI Fundamentals · Jun 2024

AWARDS & RECOGNITION

- **Special Star Award** — Outstanding Performance of the Year, Siemens (2023)
- **Star Award** — Star Performer of the Month, Siemens (Jul 2023)
- **Team Impact Award** — Best Team of the Quarter, Siemens
- **Stellar Award** — Best Employee, L&T Technology Services (2019)
- **COO Club** — Merit Club Membership, L&T Technology Services (2019–2020)

EDUCATION

B.Tech — Computer Science & Engineering

2013 – 2017

National Institute of Science & Technology, Berhampur